



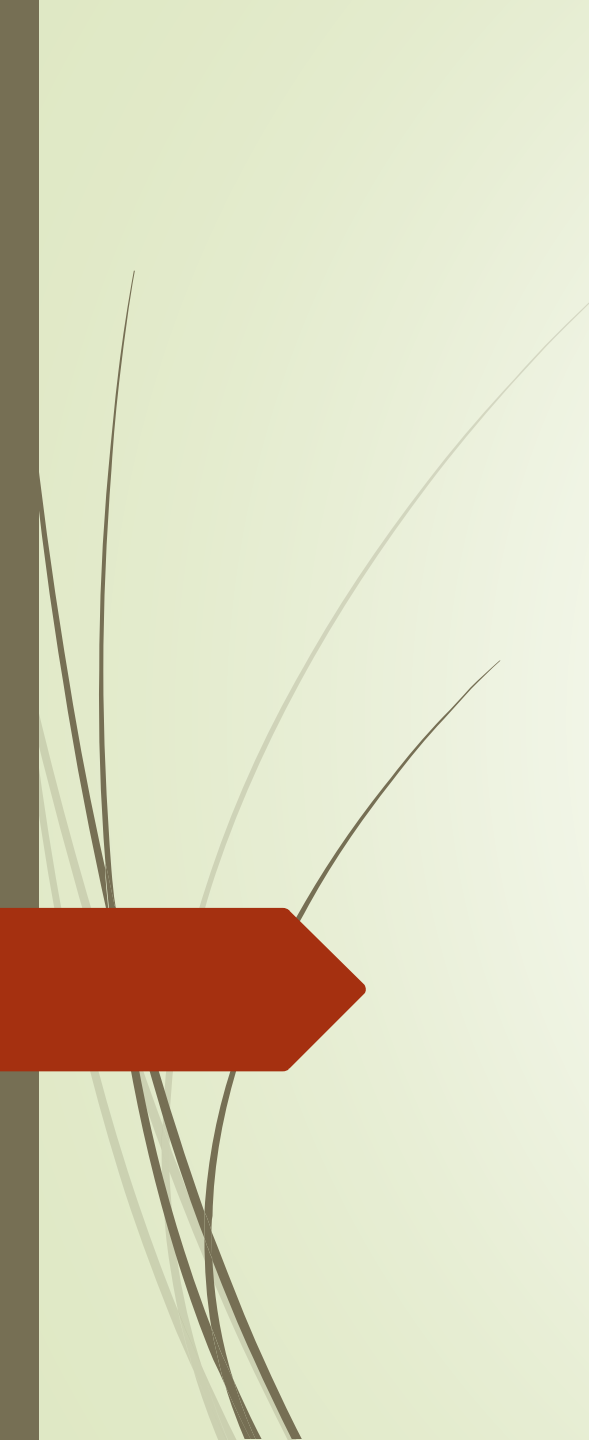
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Introduction to Robotics

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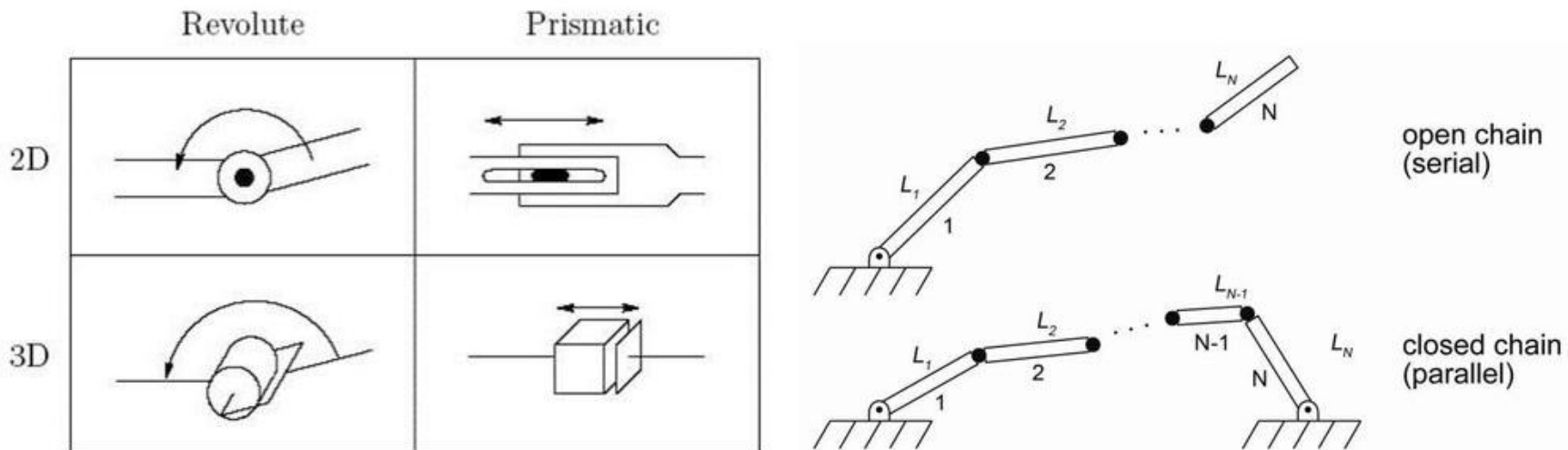
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Introduction to Robotics

Types of Robot Joints



Type	Notation	Symbol	Description
Revolute	R		Rotary motion <i>about</i> an axis
Prismatic	P		Linear motion <i>along</i> an axis

Classification of Robots

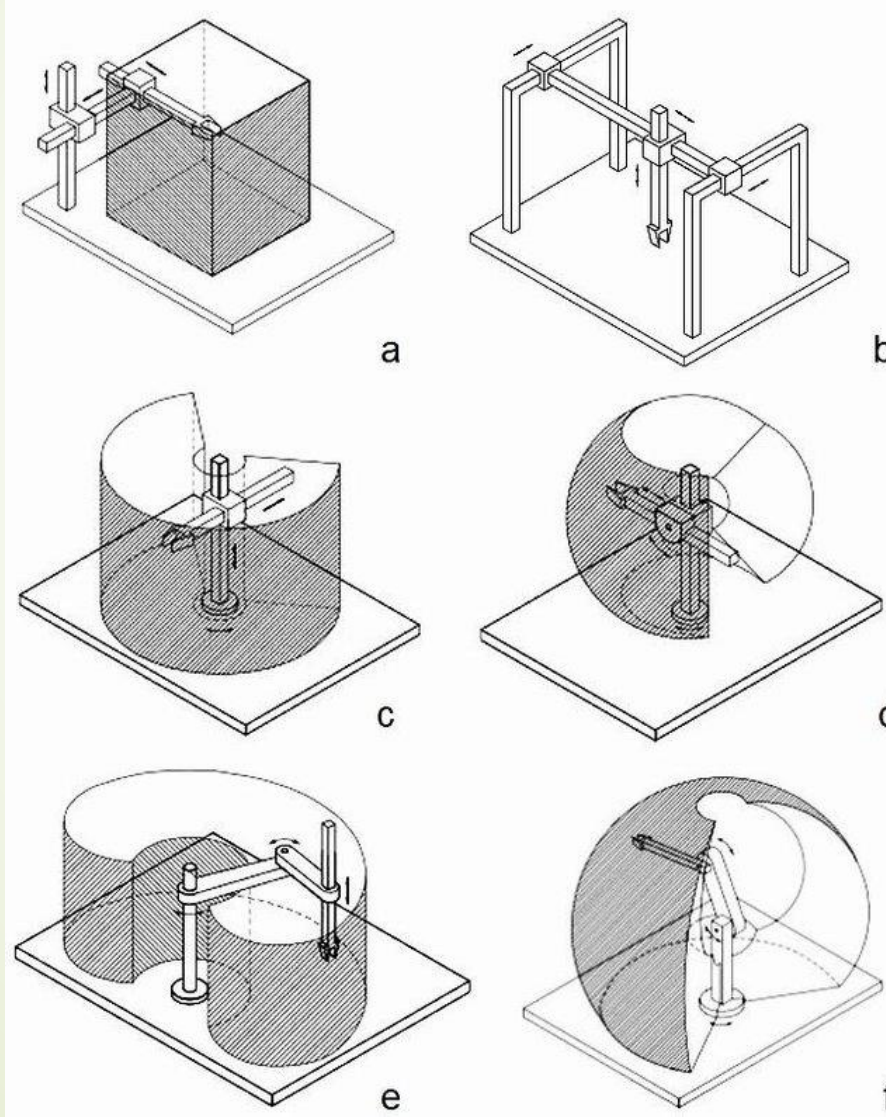
- 1. Drive Technologies such as Electric, Pneumatic and Hydraulic. Electric drives are generally DC Servomotors or DC Stepper motors.**
- 2. Work-Envelope Geometries.**
- 3. Motion Control Methods:**
 - i. Point-to-point control and,**
 - ii. Continuous path control.**

Work Envelope Geometries based on Major Axes

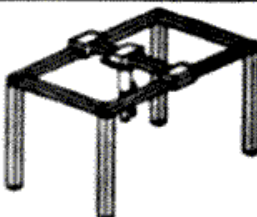
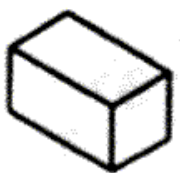
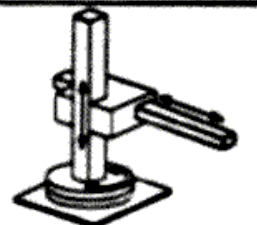
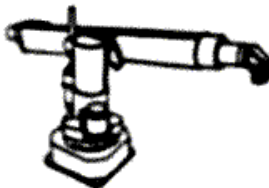
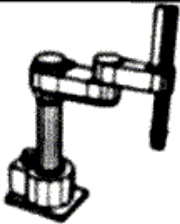
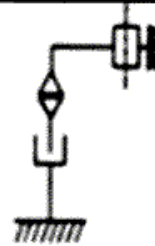
Robot	Axis 1	Axis 2	Axis 3	Total revolute
Cartesian	P	P	P	0
Cylindrical	R	P	P	1
Spherical	R	R	P	2
SCARA	R	R	P	2
Articulated	R	R	R	3

P = prismatic, R = revolute.

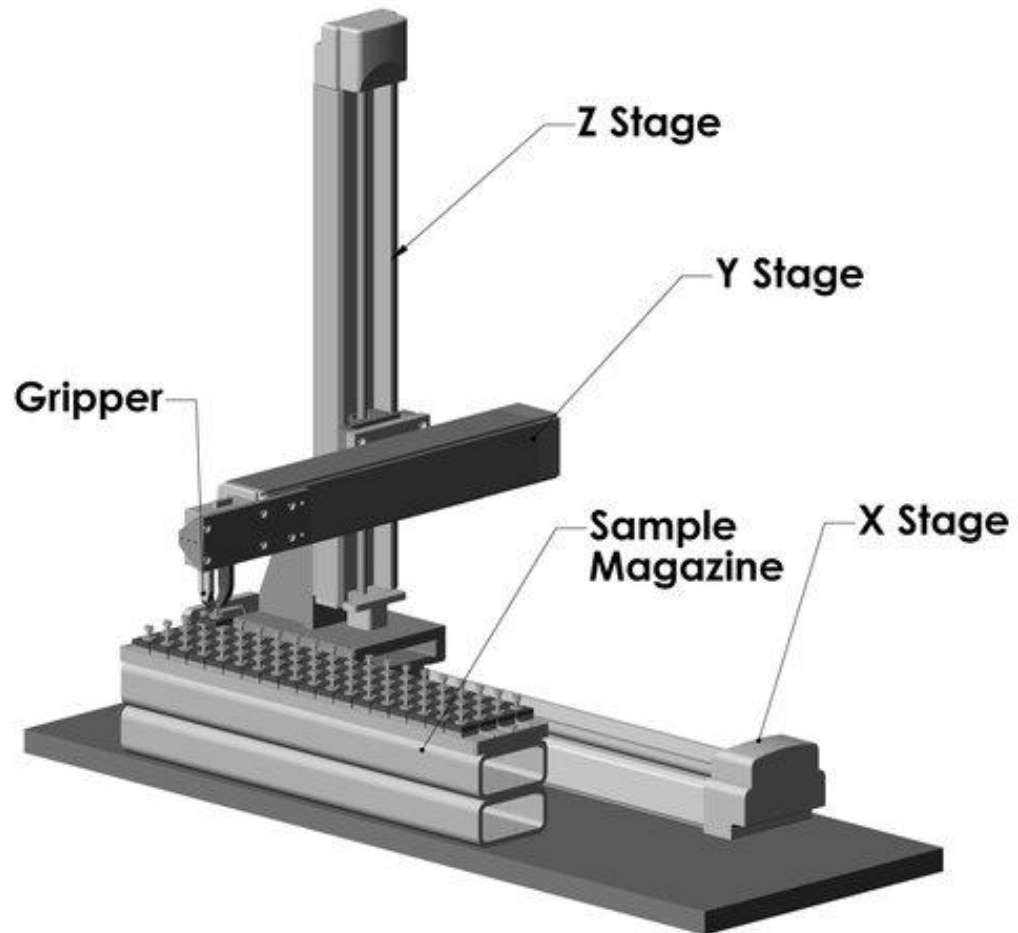
Work-Envelope Geometries (contd.)



- a. Cartesian,
x-y-z or P-P-P,
- b. Gantry,
x-y-z or P-P-P,
- c. Cylindrical,
 θ -z-r or R-P-P,
- d. Spherical,
 θ - ϕ -r or R-R-P,
- e. SCARA,
 θ - ϕ -z or R-R-P,
- f. Articulated,
 θ - ϕ - ψ or R-R-R.

Principle	Kinematic Structure	Workspace
 Cartesian Robot		
 Cylindrical Robot		
 Spherical Robot		
 SCARA Robot		
 Articulated Robot		

Cartesian Robotic Manipulator (P-P-P)



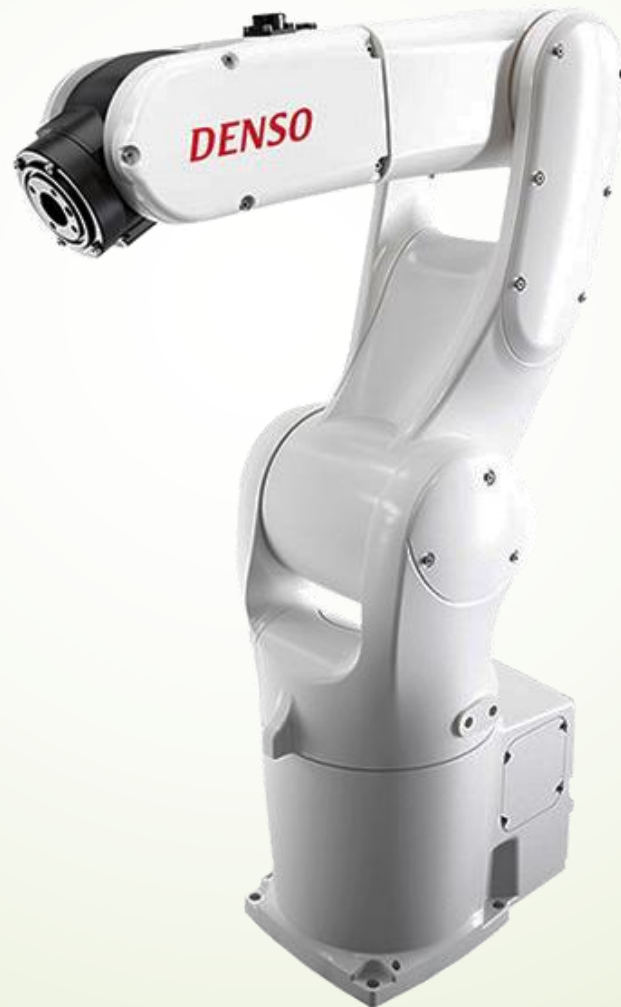
Gantry Robotic Manipulator (P-P-P)



Cylindrical Robotic Manipulator (R-P-P)



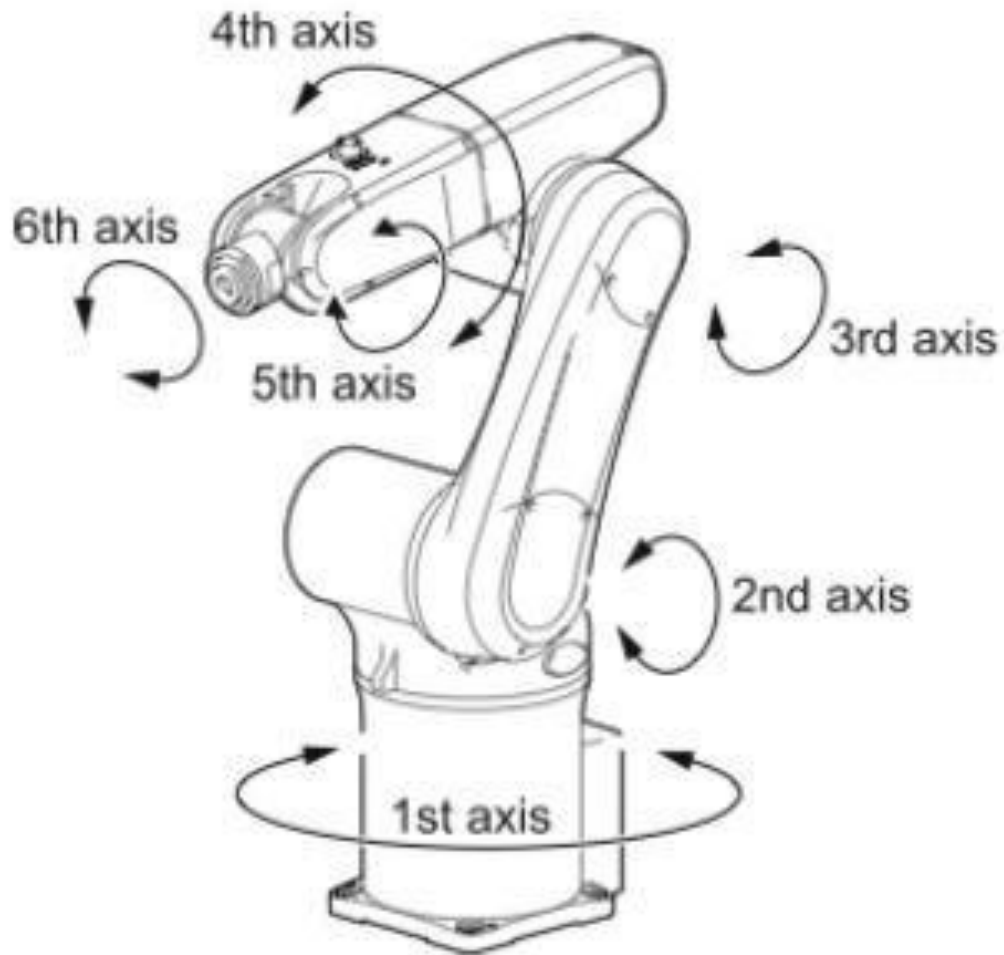
Spherical Robotic Manipulator (R-R-P)



SCARA (Selective Compliance Assembly Robot Arm) Robotic Manipulator (R-R-P)



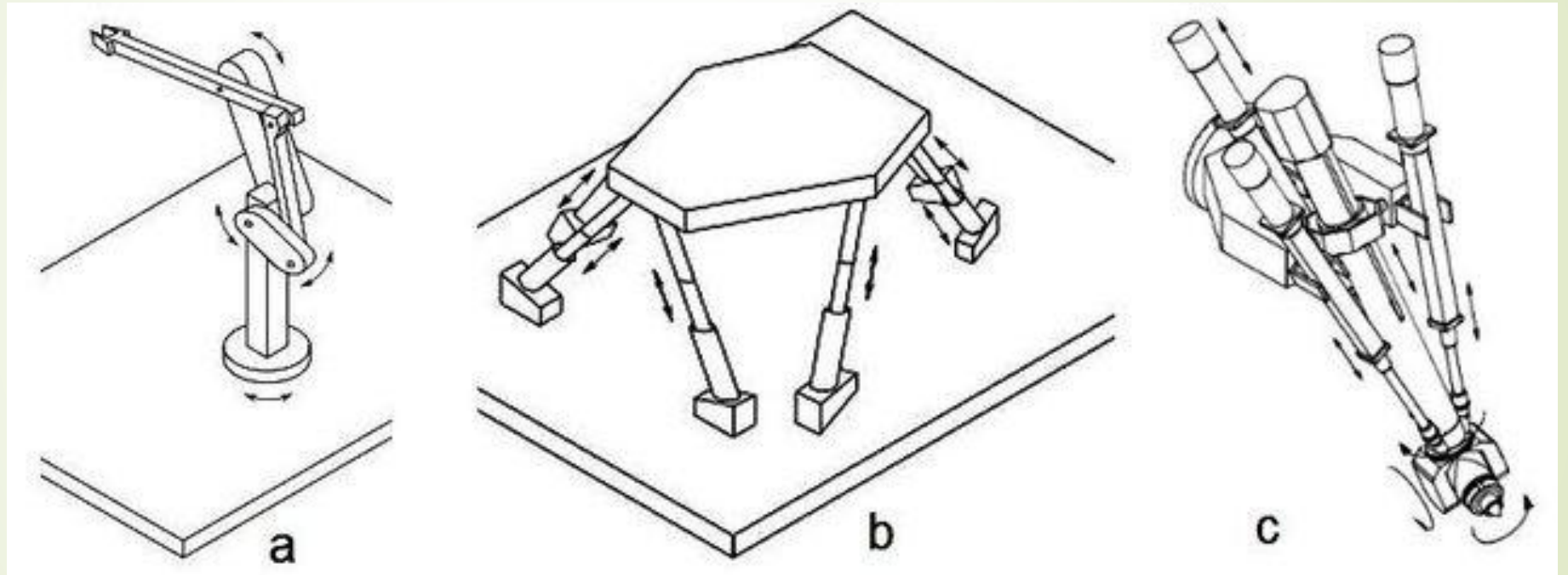
Articulated Robotic Manipulator (R-R-R)



Articulated Robotic Manipulator (Z-Y-Z Spherical Wrist)



Parallel Robotic Manipulators

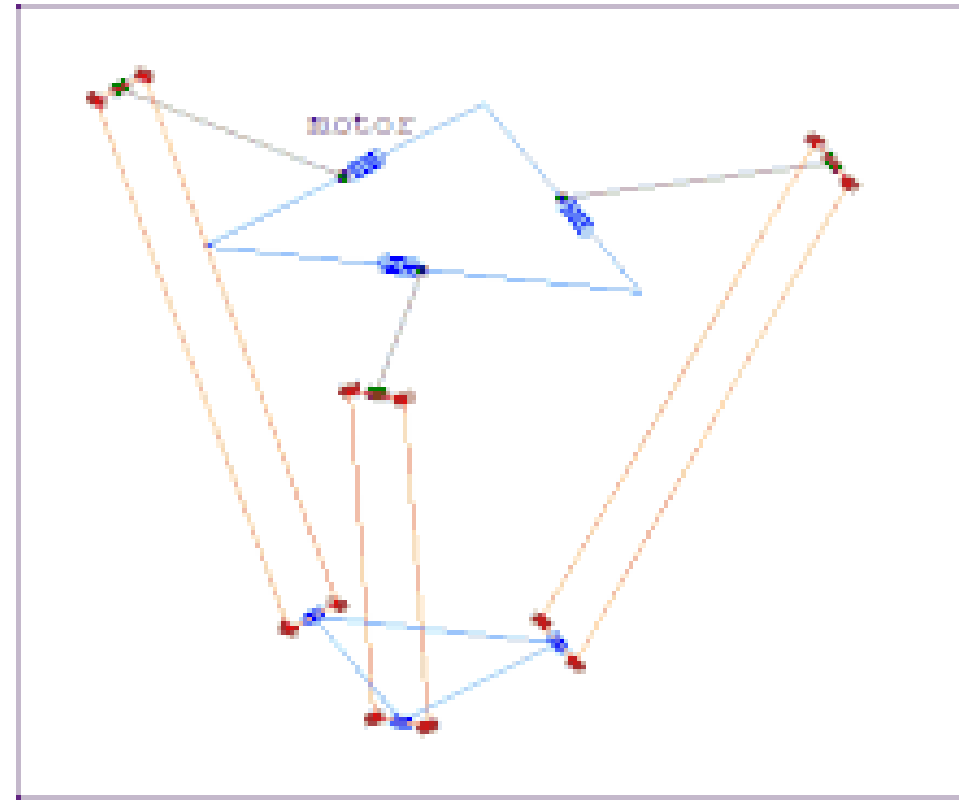


**a. Closed Chain with
Parallelogram.**

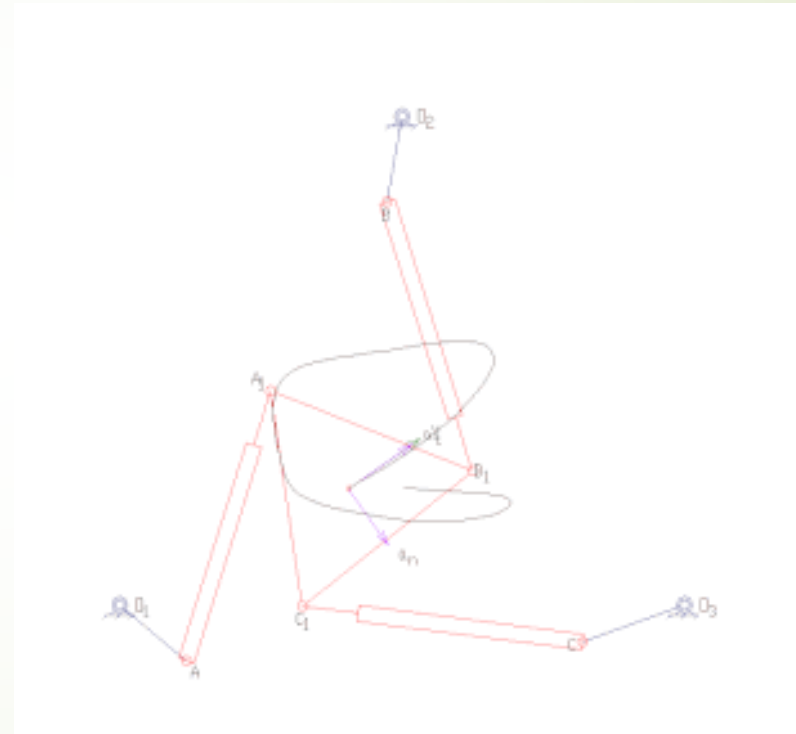
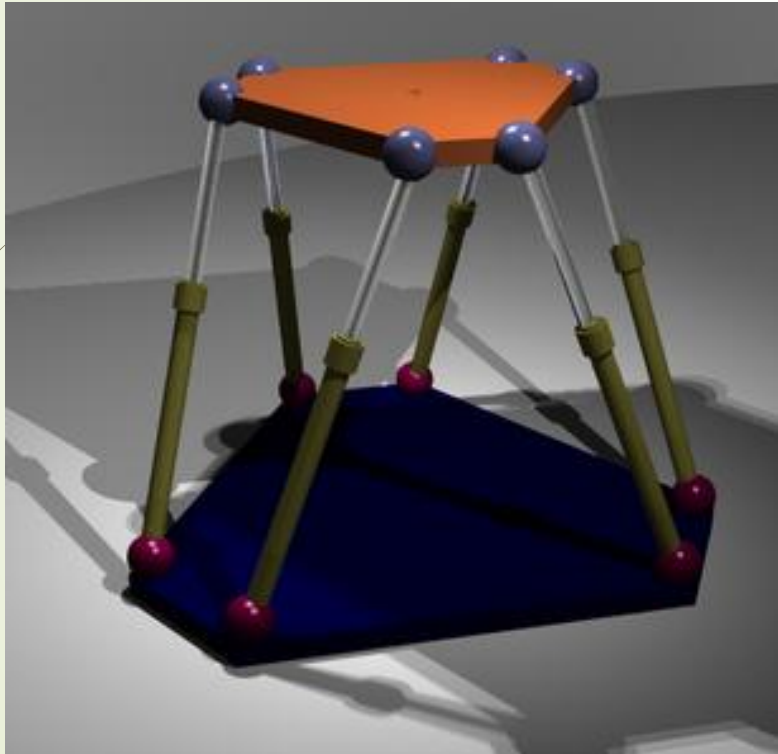
**b. Parallel.
e.g., Stewart's Platform.**

c. Hybrid Parallel-Serial.

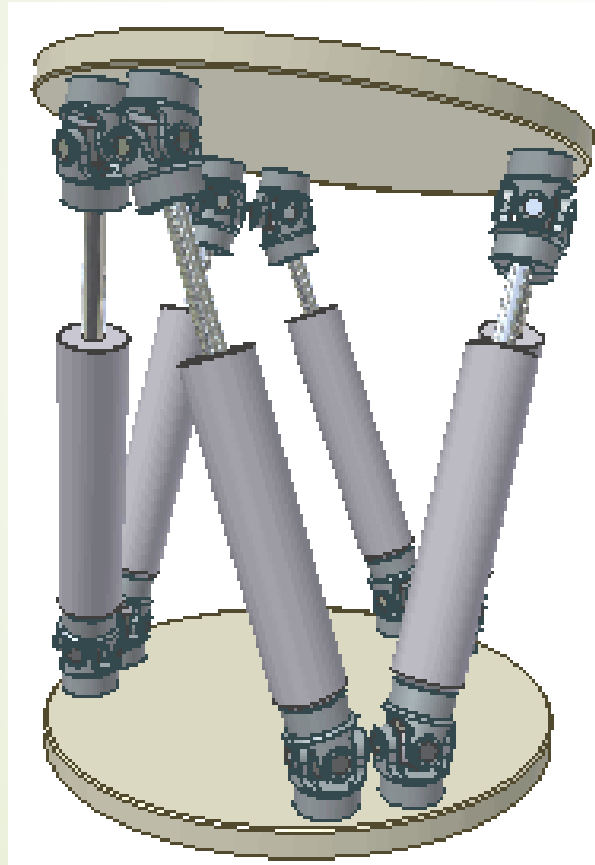
Delta Robot (Parallel Robot)



Gough-Stewart Platform (Hexapod Platform) Parallel Robot



Gough-Stewart Platform (6 DoF Platform) Parallel Robot



Application: Flight Simulator

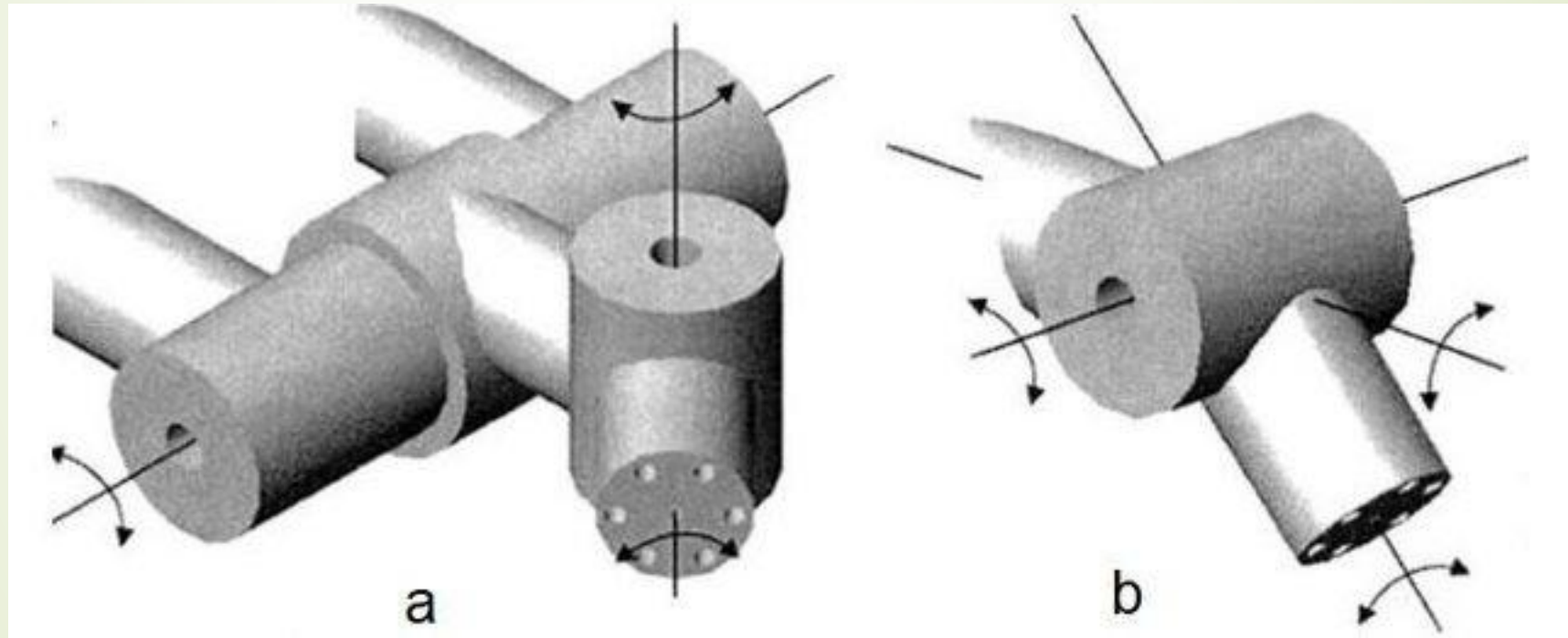


AMiBA Radio Telescope

Comparison between Parallel and Serial Manipulators

	Type of manipulator	
	Parallel manipulator	Serial manipulator
Type of manipulators	Closed loop	Open loop
End effectors	Platform	Gripper
Natural description	In Cartesian space	In joint space
Location of actuators	Near the immobile base	On the links
Inertia forces & stiffness	Less and high respectively	High and less respectively
Design considerations	Structure, workspace considerations, singularities, link interference	Strength and stiffness considerations, vibration characteristics.
Preferred property	Stiffness	Dexterity
Use of direct kinematics	Difficult and complex	Straightforward and unique
Use of inverse kinematics	Straightforward and unique	Complicated
Singularity	Static	Kinematic
Direct force transformation	Well defined and unique	Not well defined; may be non-existent, unique or infinite
Preferred application	Precise positioning	Gross motion

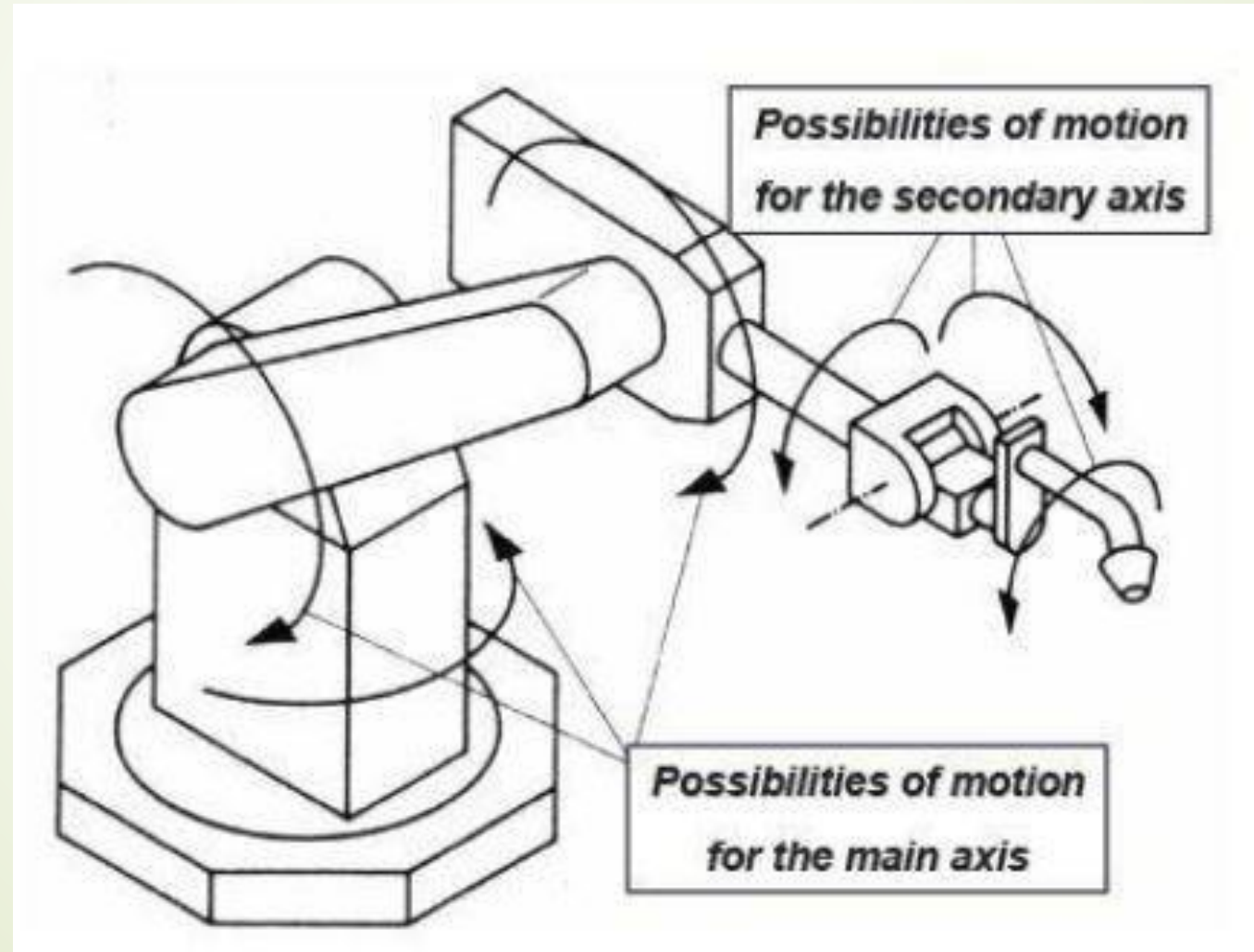
Configuration of Spherical Wrist Designs



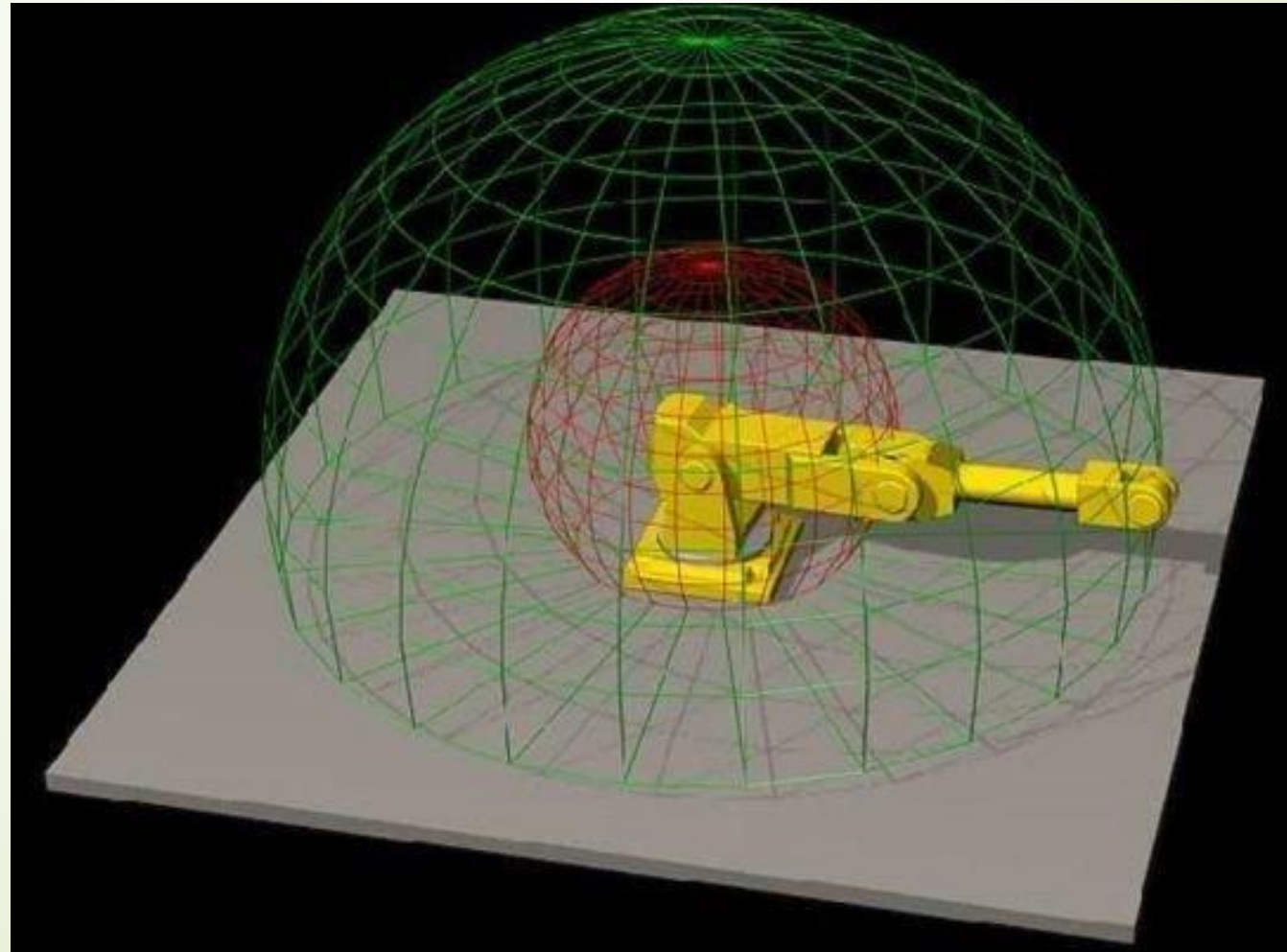
a. pitch-yaw-roll (Y-X-Z)

b. roll-pitch-roll (Z-Y-Z)

Robotic Arm with a Spherical Wrist



Main operating area and secondary operating area for a robotic arm with a spherical wrist

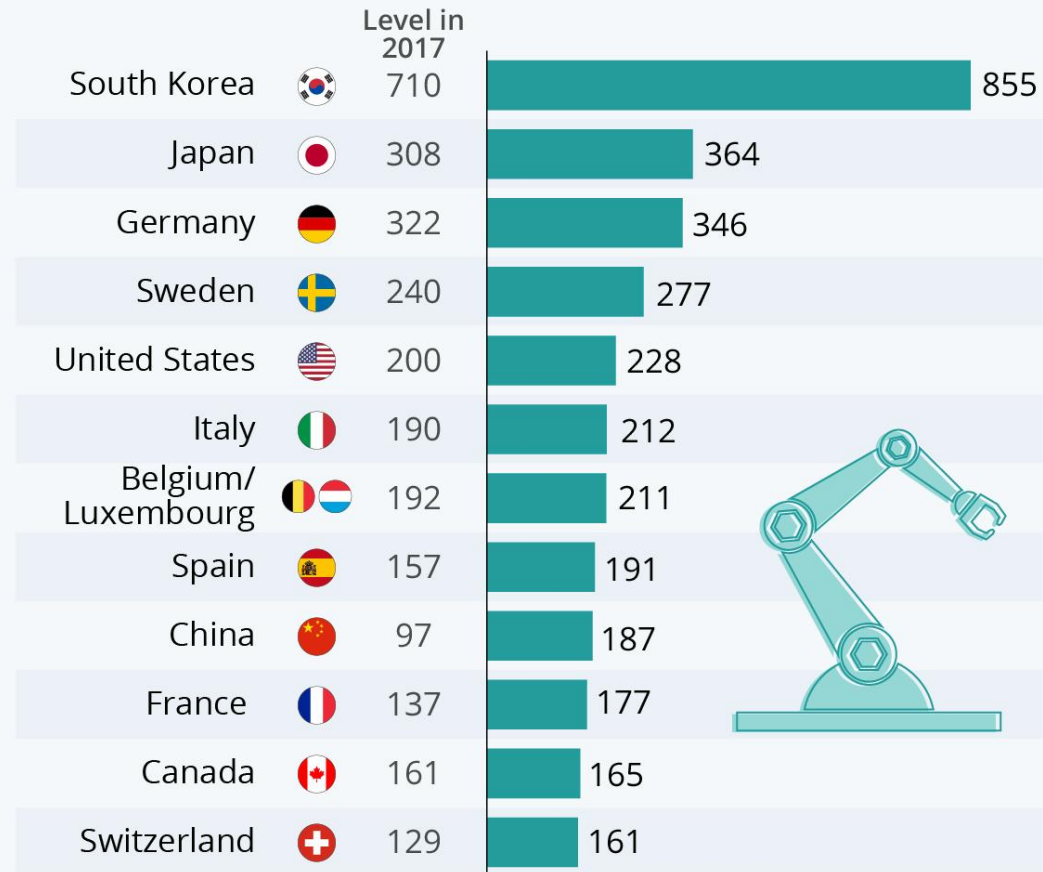


Robot Motion Control Applications

Control method	Applications
Point to point	Spot welding Pick-and-place Loading and unloading
Continuous path	Spray painting Arc welding Gluing

The Countries With The Highest Density Of Robot Workers

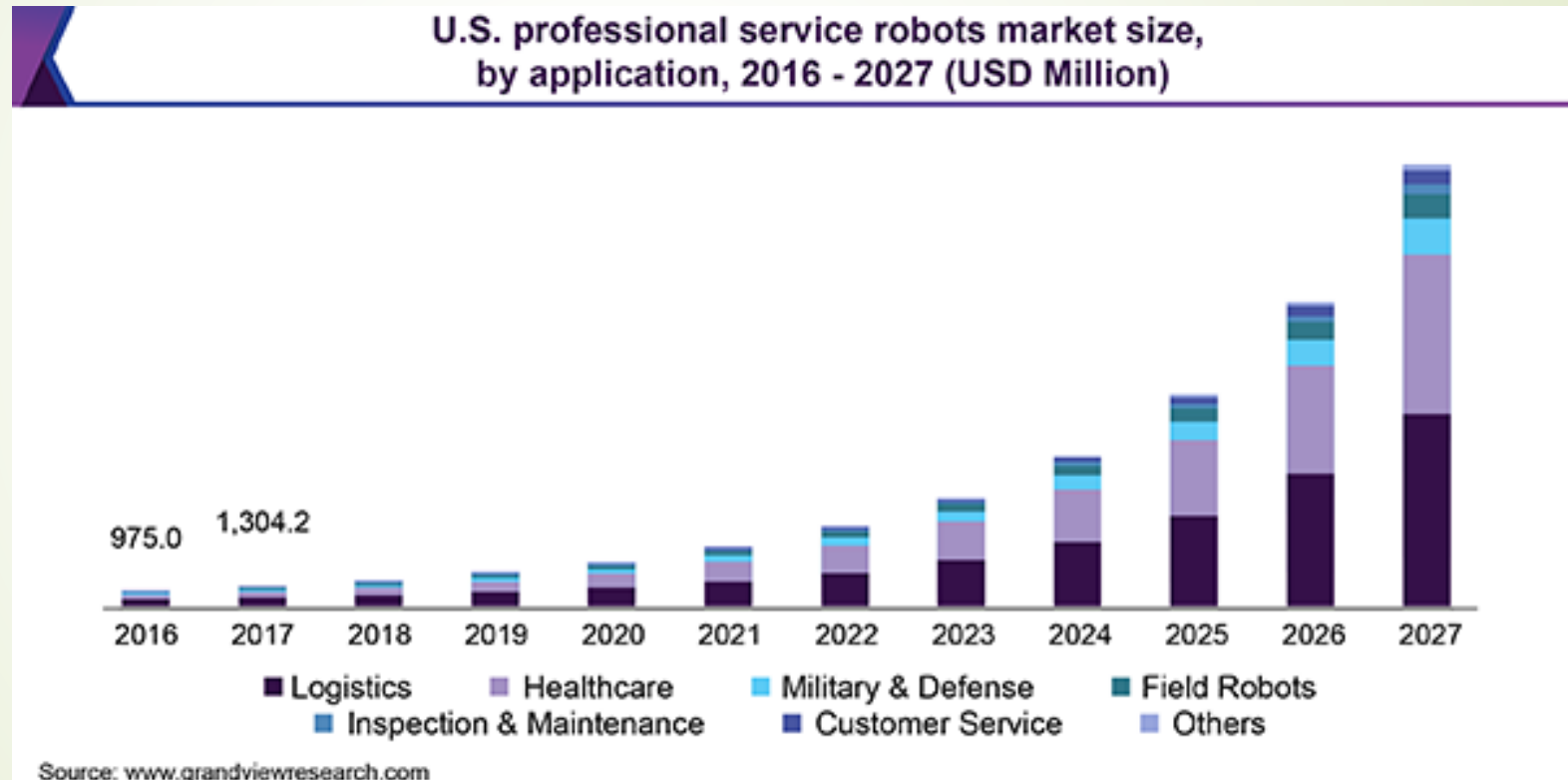
Installed industrial robots per 10,000 employees in the manufacturing industry in 2019*



* Selected countries

Source: International Federation of Robotics

Robot Applications (valued at \$ 12.3 billion globally), 2019



Applications of Industrial Robots

INDUSTRIAL ROBOTS

